

Chapman Conference on the Earth's radiation belts and inner magnetosphere

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Late in the evening on 31 January 1958, a Juno (Jupiter-C) rocket blasted into space, lofting the first U.S. artificial Earth satellite into orbit. This spacecraft, dubbed Explorer 1, joined in space one other satellite, Sputnik 2, which had been launched on 3 November 1957 by the Soviet Union. The Explorer 1 mission was groundbreaking, for it carried a small scientific payload prepared at the University of Iowa by a team of researchers led by James A. Van Allen. The instrumentation on Explorer 1 (and on the subsequently successful Explorer 3) would make the first truly revolutionary discovery of the space age, namely, that Earth is enshrouded in toroids, or belts, of extraordinarily high energy, high-intensity radiation.

Van Allen and his team would go on to explore and map the radiation regions around the Earth in considerable detail. The radiation zones are today known in his honor as the Van Allen radiation belts. Van Allen would lead subsequent exploratory efforts (such as those on Pioneer 10 and 11) that would discover radiation zones around the giant magnetized planets Jupiter and Saturn. Moreover, Van Allen and colleagues would show the lack of persistent radiation belts around Mars and Venus, owing to these latter planets' lack of strong intrinsic magnetic dipole fields.

Since the seminal discovery by Van Allen and coworkers, the world scientific community has made huge strides not only in understanding the fundamental and cosmic significance of the radiation first measured in 1958 but also in recognizing its significance as a space weather threat to human technological systems in space. At an AGU Chapman Conference held 17–22 July 2011 in St. John's, Newfoundland, Canada, more than 110 researchers from more than 15 countries gathered to assess the state of radiation belt research. Despite more than 5 decades of radiation belt research, the Chapman Conference, entitled “Dynamics of the Earth's Radiation Belts and Inner Magnetosphere,” revealed a still vibrant interest in the particles, waves, and fields that form the physical basis of modern radiation belt science.

The meeting was the first Chapman Conference dealing with space science topics to be held in Canada in more than 35 years. The meeting took place overlooking the harbor of St. John's and the beautiful narrows to the Atlantic Ocean, at a location within sight of Signal Hill, where Guglielmo Marconi first successfully received transatlantic wireless radio transmissions from Cornwall, England, in December 1901. The rugged coastline of Newfoundland provided a stunning backdrop for conference presentations and for the superb social program organized under the aegis of the Memorial University of Newfoundland. As the third nation in space, with the launch of the Alouette 1 satellite on 29 September 1962 and its 10 years of operation, Canada has a long heritage of space exploration. Hence, it was fitting that current advances in space radiation research be showcased in Canada; strong Canadian participation highlighted how Canadian space science and space weather research continues to play a major role in coordinated international endeavors.

The program of plenary lectures, invited talks, contributed oral papers, and outstanding posters showed the exciting state of modern radiation belt research. Continuing analysis of data (see Figure 1) from such Earth-orbiting spacecraft as the Combined Release and Radiation Effects Satellite (CRRES), the Solar Anomalous and Magnetospheric Particle Explorer (SAMPEX), Polar, and other active missions, as well as from physical modeling that is increasingly thorough and sophisticated, showed that radiation belt science is thriving. All eyes at the meeting were on new space missions now in development, such as the Radiation Belt Storm Probes (RBSP) dual-spacecraft project of NASA, the Energization and Radiation in Geospace (ERG) mission of Japan, and the RESONANCE mission of Russia. These new spacecraft, all planned for launch within the next 2–3 years, should further revolutionize scientific understanding of Earth's radiation belts.

The St. John's Chapman Conference effectively reminded all present not only of the exciting upcoming scientific op-

portunities in radiation belt research but also of the research's great practical significance. Spacecraft operating in low- and medium-altitude orbits around the Earth are constantly bombarded by Van Allen belt radiation. Numerous operational anomalies and outright failures of spacecraft systems and subsystems have been directly linked to radiation belt space weather episodes. The conference highlighted these space weather concerns and emphasized the need for better radiation belt forecasting and predictive understanding.

Current outstanding issues in radiation belt science, which were well addressed at the conference, include modeling the transport, acceleration, and loss of radiation belt

particles. Progress was reported on perfecting radiation belt models that combine the fundamental process of inward radial particle transport with local acceleration due to cyclotron resonance with whistler mode chorus. Recently, new mechanisms of electron energization have been discovered due to special forms of nonlinear phase trapping by coherent whistler mode waves that can accelerate electrons to mega-electron volt energies over much shorter time scales than those estimated by quasi-linear theory. How to appropriately incorporate such nonlinear processes in models that have traditionally relied on quasi-linear theory is a dilemma that was actively discussed at the conference.

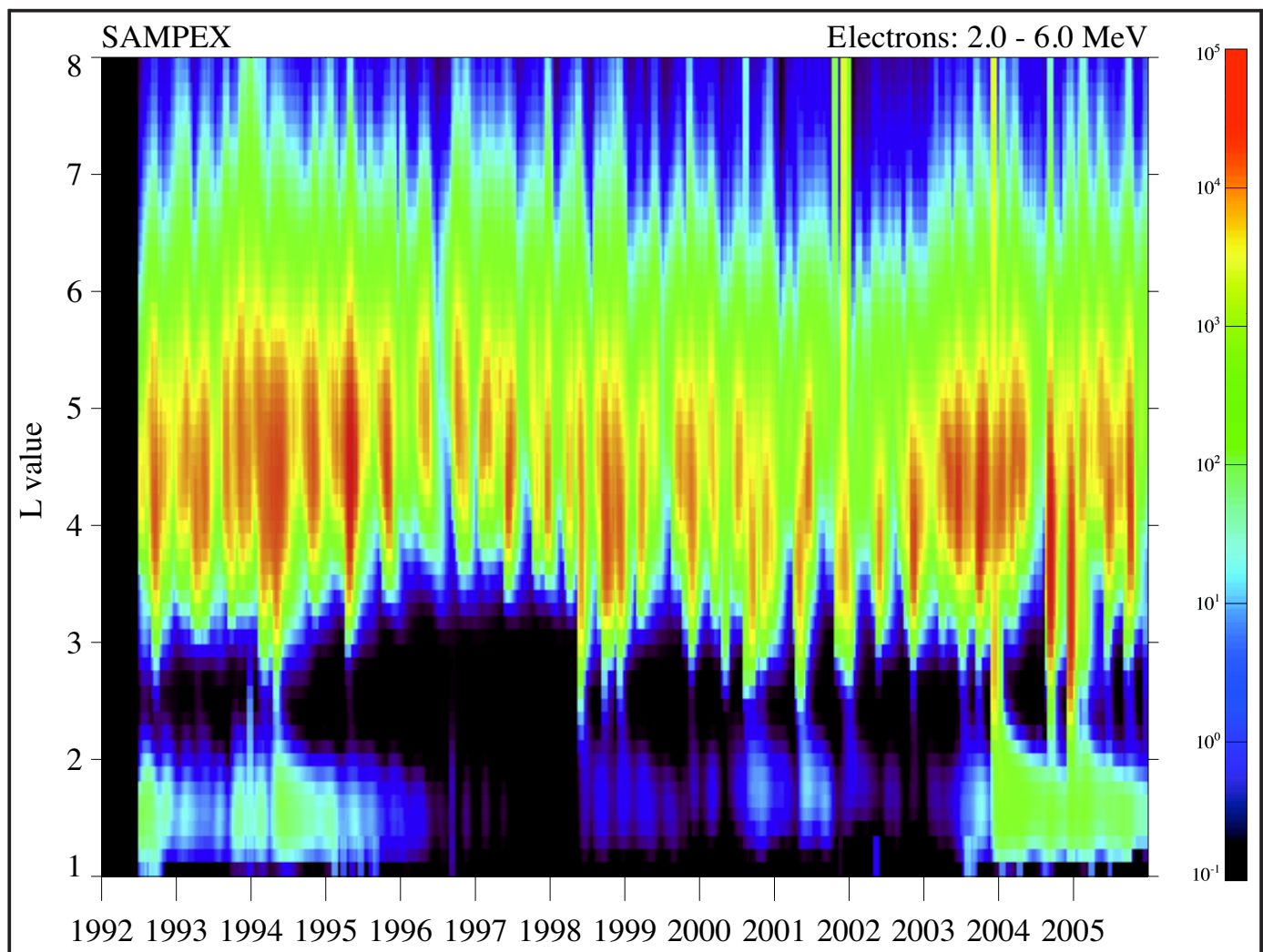


Figure 1. Solar Anomalous and Magnetospheric Particle Explorer (SAMPEX) measurements of the flux of electrons with energies of 2–6 mega-electron volts in the radiation belts from 6 July 1992 to 31 December 2005. The vertical axis is the L shell parameter, the distance in Earth radii at which a dipolar magnetic field line crosses the equatorial plane, and the color bar shows electron flux. The outer belt at $3 < L < 6$ peaks during high-speed solar wind streams in 1993–1995 and is intense in 2003–2005, including the effects of the powerful 2003 Halloween storms, during which the outer belt shifted inward, the slot region ($L < 3$) filled, and a persistent new inner belt at $L < 2$ was created. A weak inner belt is also seen in 1992–1993, created during a large March 1991 storm. At sunspot minimum (1996) the belts are weak, with electron events seen only briefly at the equinoxes. Data from 2008–2009 shown at the conference demonstrated similar behavior, with the outer radiation belt virtually disappearing during this profound minimum in solar activity.

Another topic that received much attention was the crucial role played by particle losses in controlling the variability of the radiation belts. Dramatic depletions of the radiation belts seen at geosynchronous orbit and at lower altitudes are still not well understood. Particle losses result from precipitation due to pitch angle scattering into the atmospheric loss cone by plasma waves or loss through the magnetopause. Realistic modeling of the transport, acceleration, and loss of radiation belt particles requires at least a detailed knowledge of the magnetic field configuration during disturbed conditions, as well as spectral information on the relevant plasma wave modes with respect to magnetic local time and latitude. The impending satellite missions discussed at the conference are expected to contribute greatly to providing these important data.

The more than 70 talks and 60 posters at the St. John's meeting will form the basis of a new AGU monograph on radiation belt and inner magnetosphere science. The conference conveners and authors of this report thank AGU, Memorial University of Newfoundland, the city of St. John's, the Johns Hopkins University Applied Physics Laboratory, and NASA for the organizational and financial support that made the conference a huge success. The radiation belt community now looks forward to using this conference as a springboard to greater future successes from imminent new spacecraft missions. It

will, indeed, be a fitting tribute to the pioneering efforts of Van Allen that this conference will light the way to a new level of understanding of the radiation regions that bear his name.

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